Practice workbook · Area between two curves (or a curve and a line)

Finding the area enclosed by two curves (or a curve and a line) by integrating the top minus the bottom between intersections.

11 question parts: 1 worked-example retry and 10 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-CAL-20

1. Question 5 · part c · 7 marks

2023 NOV · Paper 02

Worked-example retry · Attempt part c only.

5

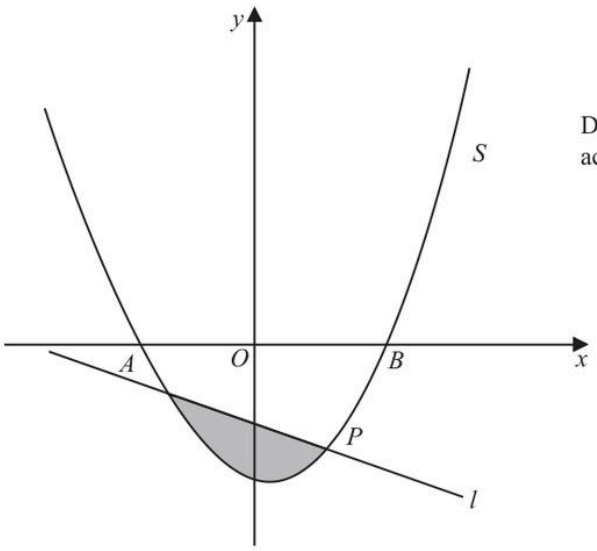


Figure 1

Figure 1 shows part of the curve S with equation $y = px^2 + qx + r$ where p, q and r are constants.

The points A, B and P with coordinates $(-2, 0)$, $(6, 0)$ and $(4, -6)$ respectively lie on S

(a) Show that an equation of S is $y = \frac{x^2}{2} - 2x - 6$

(3)

The line l is the normal to S at the point P

(b) Show that an equation of l is $2y + x + 8 = 0$

(5)

The finite region shown shaded in Figure 1 is bounded by S and l

(c) Use algebraic integration to find the exact area of the shaded region.

(7)

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1. Question 5 · part c · 7 marks

2023 NOV · Paper 02

Worked-example retry · Attempt part c only.

Question 5 continued

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1. Question 5 · part c · 7 marks

2023 NOV · Paper 02

Worked-example retry · Attempt part c only.

Question 5 continued

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1. Question 5 · part c · 7 marks

2023 NOV · Paper 02

Worked-example retry · Attempt part c only.

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Question 5 continued

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(Total for Question 5 is 15 marks)

1. Question 5 · part c · 7 marks · Working and answer

Working and answer

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2. Question 10 · part d · 4 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part d only.

10 $f(x) = 6x - x^2 \qquad x \in \mathbb{R}$

Given that $f(x)$ can be written in the form $D(x + E)^2 + F$ where D, E and F are integers,

(a) find the value of D, the value of E and the value of F.

(3)

(b) Find

(i) the maximum value of $f(x)$,

(ii) the value of x for which the maximum occurs.

(2)

The curve C has equation $y = f(x)$

The curve S has equation $y = x^2 - 4x + 8$

The curve S intersects the curve C at two points.

(c) Find the coordinates of each of these two points.

(4)

The finite region R is bounded by the curve C and the curve S.

(d) Use algebraic integration to find the area of R.

(4)

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2. Question 10 · part d · 4 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part d only.

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Question 10 continued

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2. Question 10 · part d · 4 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part d only.

Question 10 continued

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2. Question 10 · part d · 4 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part d only.

Question 10 continued

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(Total for Question 10 is 13 marks)

2. Question 10 · part d · 4 marks · Working and answer

Working and answer

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3. Question 7 · part d · 5 marks

2020 NOV · Paper 01

Further practice · Attempt part d only.

7

$$f(x) = x^2 - 9x + 14$$

Given that $f(x)$ can be written in the form $(x + a)^2 + b$, where a and b are constants,

(a) find the value of a and the value of b.

(2)

(b) Hence, or otherwise, find

(i) the minimum value of $f(x)$

(ii) the value of x for which this minimum occurs.

(2)

The curve C has equation $y = f(x)$

The line l has equation $y = x + 5$

(c) Use algebra to find the coordinates of the points of intersection of C and l.

(4)

(d) Use algebraic integration to find the exact area of the finite region bounded by C and l.

(5)

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3. Question 7 · part d · 5 marks

2020 NOV · Paper 01

Further practice · Attempt part d only.

Question 7 continued

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3. Question 7 · part d · 5 marks

2020 NOV · Paper 01

Further practice · Attempt part d only.

Question 7 continued

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3. Question 7 · part d · 5 marks

2020 NOV · Paper 01

Further practice · Attempt part d only.



Question 7 continued

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(Total for Question 7 is 13 marks)

3. Question 7 · part d · 5 marks · Working and answer

Working and answer

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4. Question 10 · part b · 9 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

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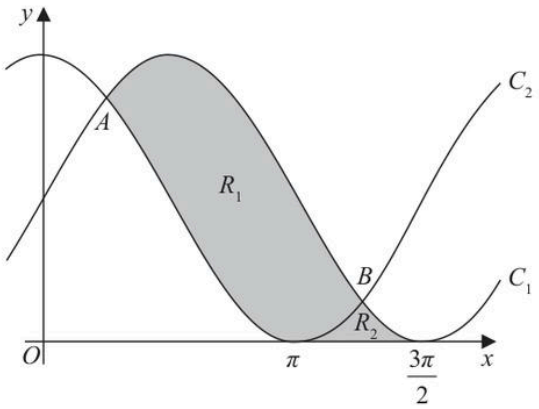


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows part of the curve C_1 with equation $y = \sin x + 1$ and part of the curve C_2 with equation $y = \cos x + 1$

As shown in Figure 2, C_1 and C_2 intersect at the point A and at the point B

(a) Find the exact value of the x coordinate of A and the exact value of the x coordinate of B

(3)

The shaded finite region R_1 shown in Figure 2 is bounded by C_1 and C_2

The shaded finite region R_2 shown in Figure 2 is bounded by the x-axis, C_1 and C_2

(b) Use calculus to find the ratio

area of R_1 : area of R_2

Give your answer in the form $a : \left(\frac{\pi\sqrt{2}}{b} - c\right)$ where a, b and c are integers.

(9)

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4. Question 10 · part b · 9 marks

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Further practice · Attempt part b only.

Question 10 continued

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4. Question 10 · part b · 9 marks

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Further practice · Attempt part b only.

Question 10 continued

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4. Question 10 · part b · 9 marks

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Further practice · Attempt part b only.

Question 10 continued

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(Total for Question 10 is 12 marks)

4. Question 10 · part b · 9 marks · Working and answer

Working and answer

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5. Question 11 · part c · 8 marks

2021 JUN · Paper 01

Further practice · Attempt part c only.

11 (a) Using a formula from page 2, show that $\cos 2x = 1 - 2\sin^2 x$

(3)

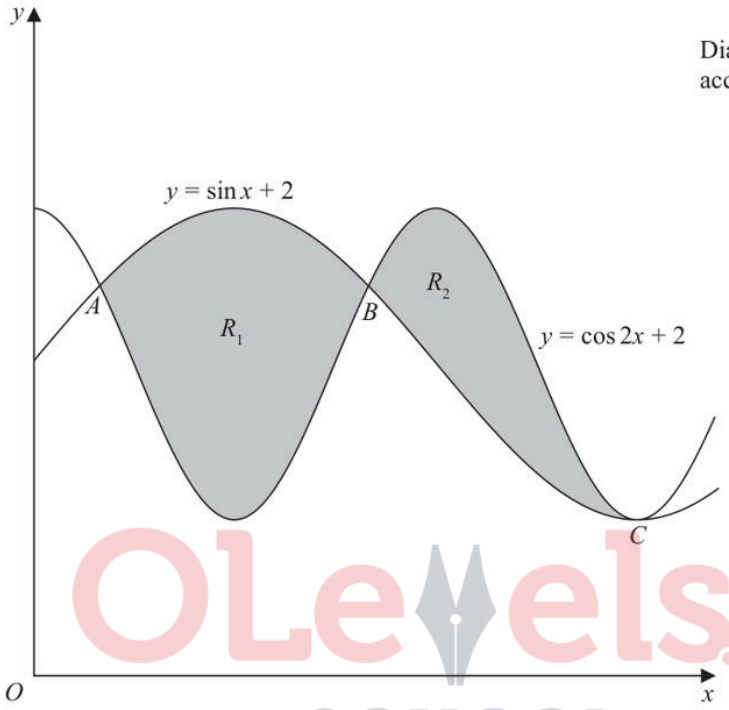


Figure 2

Figure 2 shows a sketch of part of the curves with equations $y = \sin x + 2$ and $y = \cos 2x + 2$

The points A, B and C, shown in Figure 2, are three points that are common to both curves.

(b) Find the coordinates of each of these points.

(4)

R_1 and R_2, shown shaded in Figure 2, are two regions enclosed by the two curves.

(c) Use calculus to find, in its simplest form, the ratio

$$\text{area of } R_1 : \text{area of } R_2$$

(8)

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5. Question 11 · part c · 8 marks

2021 JUN · Paper 01

Further practice · Attempt part c only.

Question 11 continued

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5. Question 11 · part c · 8 marks

2021 JUN · Paper 01

Further practice · Attempt part c only.

Question 11 continued

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5. Question 11 · part c · 8 marks

2021 JUN · Paper 01

Further practice · Attempt part c only.

Question 11 continued

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5. Question 11 · part c · 8 marks

2021 JUN · Paper 01

Further practice · Attempt part c only.

Question 11 continued

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(Total for Question 11 is 15 marks)

5. Question 11 · part c · 8 marks · Working and answer

Working and answer

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6. Question 9 · part whole · 8 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part whole only.

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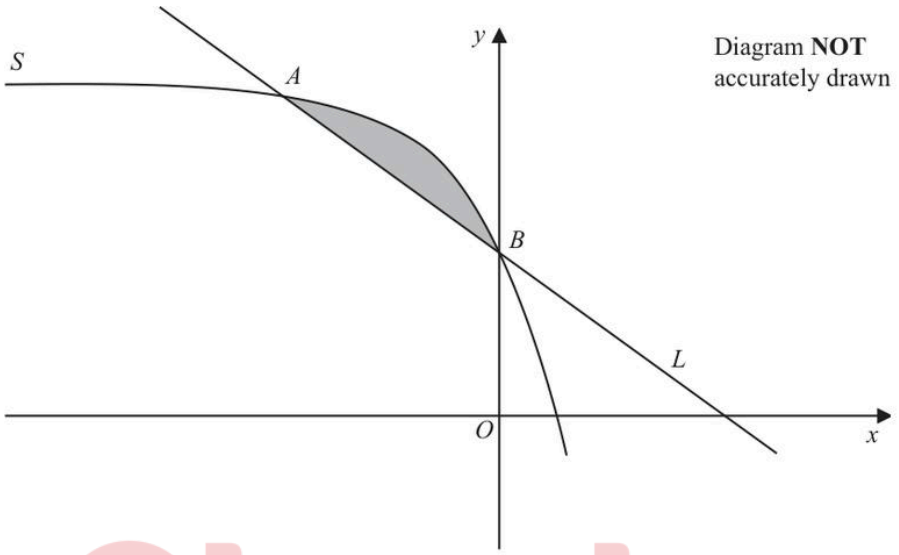


Figure 3

Figure 3 shows a sketch of part of the curve S with equation $y = -2e^{3x} + 4$ and the line L

The curve S has intersections with the line L at the points A and B with x coordinates $x = -1$ and $x = 0$ respectively.

The finite region bounded by S and L is shown shaded in Figure 3

Use calculus to find the exact area of this region.

Give your answer in the form $\frac{a + be^{-c}}{c}$ where a, b and c are integers to be found.

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6. Question 9 · part whole · 8 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part whole only.

Question 9 continued

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6. Question 9 · part whole · 8 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part whole only.

Question 9 continued

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6. Question 9 · part whole · 8 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part whole only.

Question 9 continued

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(Total for Question 9 is 8 marks)

6. Question 9 · part whole · 8 marks · Working and answer

Working and answer

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7. Question 9 · part b · 7 marks

2022 JUN · Paper 01

Further practice · Attempt part b only.

9 $f(x) = 3x^4 + 4x^3 - 36x^2 + 64$

Given that $f(x)$ can be written in the form $(x-2)^2(ax^2 + bx + c)$

(a) find the value of a, the value of b and the value of c

(4)

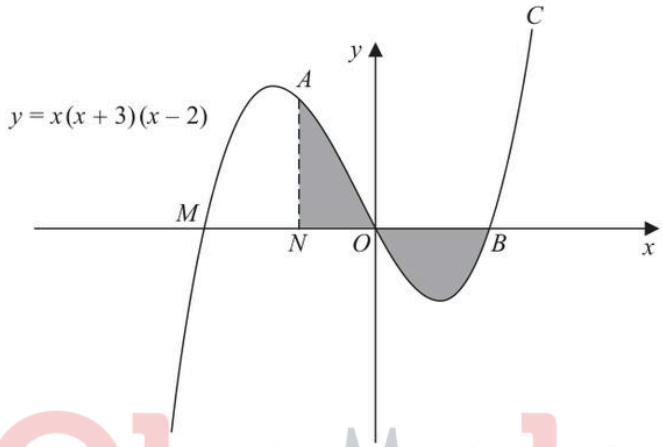


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows a sketch of part of the curve C with equation $y = x(x+3)(x-2)$

The curve C crosses the x-axis at the point M, the origin and the point B.
The point N lies on the x-axis between M and O.
The point A lies on C such that AN is parallel to the y-axis.

The area of the shaded region bounded by the curve and OB is numerically equal to the area of the shaded region bounded by the curve, ON and NA.

Given that the coordinates of N are $(n, 0)$,

(b) use algebraic integration to show that n satisfies the equation

$$(x-2)^2(3x^2 + 16x + 16) = 0$$

(7)

(c) Hence find the exact coordinates of A.

(5)

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7. Question 9 · part b · 7 marks · Working and answer

Working and answer

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8. Question 11 · part d · 5 marks

2023 JUN · Paper 01

Further practice · Attempt part d only.

11 $f(x) = 10 + 6x - x^2$

Given that $f(x)$ can be written in the form $A(x + B)^2 + C$ where A, B and C are constants,

(a) find the value of A, the value of B and the value of C

(4)

(b) Hence, or otherwise, find

(i) the value of x for which $f(x)$ has its greatest value

(ii) the greatest value of $f(x)$

(2)

The curve C has equation $y = f(x)$

The curve S with equation $y = x^2 - x + 13$ intersects curve C at two points.

(c) Find the x coordinate of each of these two points.

(3)

(d) Use algebraic integration to find the exact area of the finite region bounded by the curve C and the curve S.

(5)

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8. Question 11 · part d · 5 marks

2023 JUN · Paper 01

Further practice · Attempt part d only.

Question 11 continued

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8. Question 11 · part d · 5 marks

2023 JUN · Paper 01

Further practice · Attempt part d only.

Question 11 continued

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(Total for Question 11 is 14 marks)

8. Question 11 · part d · 5 marks · Working and answer

Working and answer

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9. Question 10 · part b · 6 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

10 The equation of a curve C is $y = f(x)$ where $f'(x) = 3x^2 - 4x - p$ and $p \neq 0$

The points with coordinates $(2, 0)$ and $(-1, 9)$ lie on C.

(a) Show that C has equation $y = x^3 - 2x^2 - 4x + 8$

(6)

The straight line l has equation $y = 8 - 4x$

(b) Use algebraic integration to find the exact area of the finite region bounded by C and l.

(6)

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9. Question 10 · part b · 6 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 10 continued

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9. Question 10 · part b · 6 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 10 continued

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9. Question 10 · part b · 6 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 10 continued

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(Total for Question 10 is 12 marks)

9. Question 10 · part b · 6 marks · Working and answer

Working and answer

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10. Question 8 · part b · 6 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

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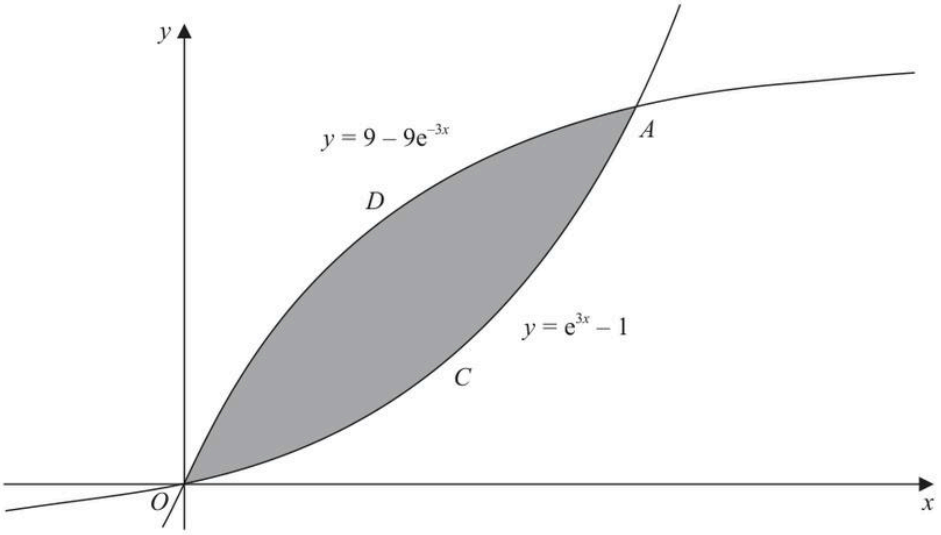


Figure 2

Figure 2 shows part of the curve C with equation $y = e^{3x} - 1$ and part of the curve D with equation $y = 9 - 9e^{-3x}$

The curves intersect at the origin O and the point A.

(a) (i) Show that the x coordinate of the point A satisfies the equation

$$(e^{3x})^2 - 10e^{3x} + 9 = 0$$

(ii) Hence, show that the x coordinate of the point A is $\frac{1}{3}\ln 9$

(5)

The finite region bounded by C and by D is shown shaded in Figure 2.

(b) Use calculus to find the exact area of this region.

(6)

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10. Question 8 · part b · 6 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 8 continued

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10. Question 8 · part b · 6 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 8 continued

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10. Question 8 · part b · 6 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 8 continued

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(Total for Question 8 is 11 marks)

10. Question 8 · part b · 6 marks · Working and answer

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11. Question 8 · part c · 5 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

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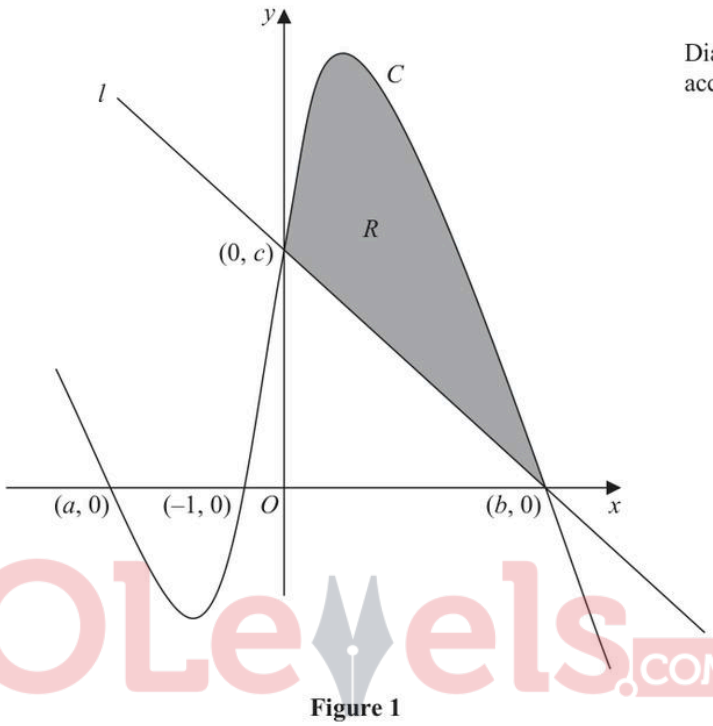


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows part of the curve C with equation $y = f(x)$

The curve C passes through the points with coordinates

$$(a, 0), (-1, 0), (b, 0) \text{ and } (0, c)$$

Given that $f'(x) = 17 + 2x - 3x^2$

(a) show that the equation of C is $y = 15 + 17x + x^2 - x^3$

(4)

(b) Find the value of a, the value of b and the value of c

(6)

The straight line l intersects C at the points with coordinates $(b, 0)$ and $(0, c)$

The region R, shown shaded in Figure 1, is bounded by l and C

(c) Use algebraic integration to find the exact area of region R

(5)

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11. Question 8 · part c · 5 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 8 continued

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11. Question 8 · part c · 5 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 8 continued

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11. Question 8 · part c · 5 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 8 continued

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(Total for Question 8 is 15 marks)

11. Question 8 · part c · 5 marks · Working and answer

Working and answer

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